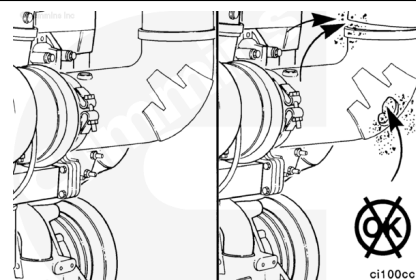


Trainings ▾

Maintenance Check

⚠ CAUTION ⚠

Engine intake air must be filtered to prevent dirt and debris from entering the engine. If the intake air piping is damaged or loose, unfiltered air will enter the engine and cause premature wear.



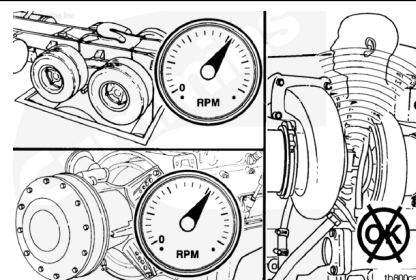
Inspect for loose clamps or damage between the intake air piping, air cleaner, turbocharger, charge-air cooler, and intake manifold.

Replace any damaged pipes, and tighten loose clamps.

Check for corrosion of the intake system piping under the clamps and hoses. Corrosion can allow corrosive products and dirt to enter the intake system. Disassemble and clean as required.

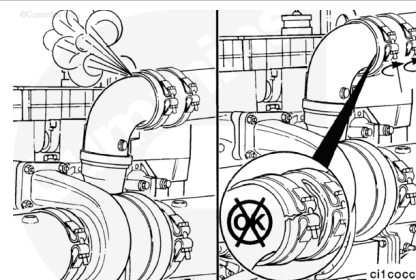
Excessive back pressure can cause exhaust leaks. Verify the exhaust back pressure is within specified limits. Refer to Procedure 011-009 in Section 11 (</qs3/pubsys2/xml/en/procedures/101/101-011-009-tr.html>).

Operate the engine at full throttle and maximum load, and check for air leaks. Listen for a whistling noise caused by high-pressure air leaks.



⚠ CAUTION ⚠

Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be able to be used.



The noise can be caused by an air leak from the following:

Turbocharger-to-charge-air cooler elbow connection.

- Inspect the connection and o-ring seal for damage.

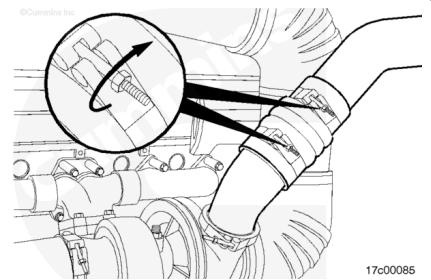
- Tighten the V-band clamps.

Torque Value: 9 n•m [80 in-lb]

Any charge-air cooler piping or connecting hose.

- Inspect the hose and piping for damage.
- Tighten the hose clamps.

Torque Value: 9 n•m [80 in-lb]

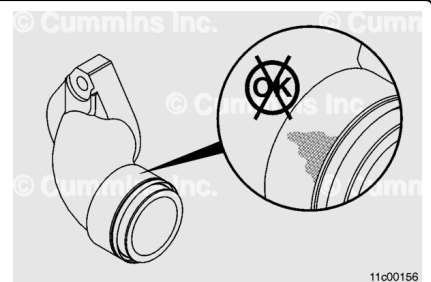


Check soot streaks from the exhaust manifold slip joints.

Remove the exhaust manifold and inspect the slip joints for damage.

Check exhaust manifold-to-cylinder head sealing gasket.

If soot streaks are found replace the exhaust manifold to cylinder head gaskets. Refer to Procedure 011-007 in Section 11 (/qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html).

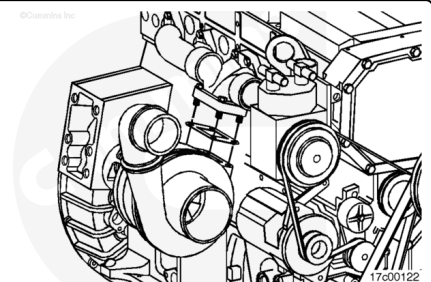


Turbocharger-to-exhaust-manifold mounting gasket.

- Replace the gasket.

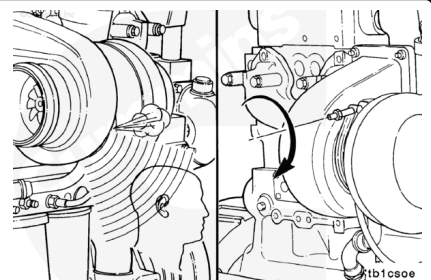
Refer to Procedure 010-033

(/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).



⚠ CAUTION ⚠

Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be able to be used.



Turbine housing sealing surface air leak.

- Tighten the V-band clamp.

Torque Value: 14 n•m [124 in-lb]

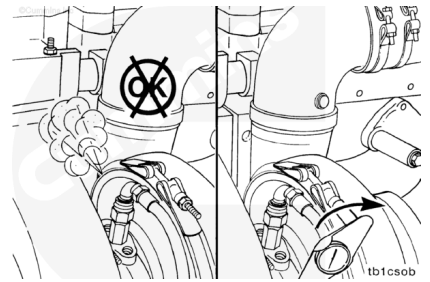
- Check for an air leak.
- If an air leak is still present, remove and replace the turbocharger.

Compressor housing sealing surface air leak.

- Tighten the V-band.

Torque Value: 9 n•m [80 in-lb]

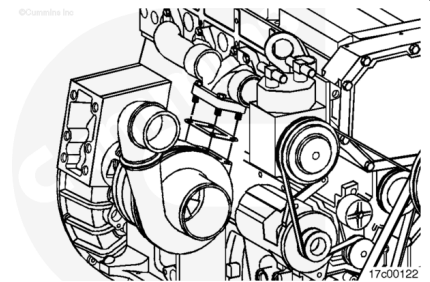
- Check for an air leak.
- If an air leak is still present, remove and replace the turbocharger. Refer to Procedure 010-033 in Section 10 (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>).



Variable Geometry

⚠ WARNING ⚠

Removing the turbine housing V-band tamper proofing on any variable geometry turbocharger will void the Cummins Inc. warranty coverage.



Turbine housing sealing surface air leak: For variable geometry turbochargers, the turbine V-band nut has a tamper proof clamp. Therefore the torque specifications for the turbine V-band will **not** be able to be checked.

The purpose of the tamper proofing is to prevent disturbing this joint. Any loosening of the V-band or adjusting of the orientation of the turbine housing will affect the sliding nozzle clearances and will damage the turbocharger and result in poor performance.

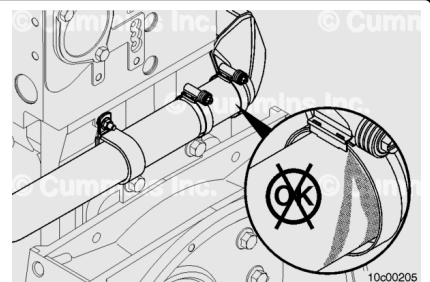
Indications of leakage are soot streaks from the joint and/or an audible whistle from the engine. The turbochargers **must** be replaced if there are any indications of leakage from the turbine housing V-band joint.

EGR plumbing sealing joints leak:

Check for soot streaks from the EGR crossover tube and hoses.

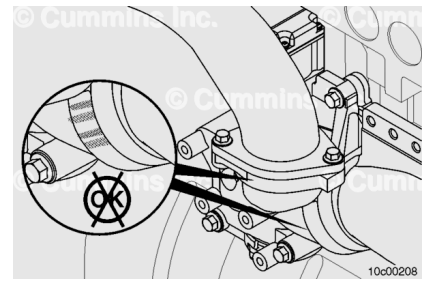
Tighten the hose clamps. Refer to Procedure 011-025 in Section 11 (</qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html>).

Check for an exhaust leak. If an exhaust leak is still present, remove and replace the EGR crossover tube or hose.



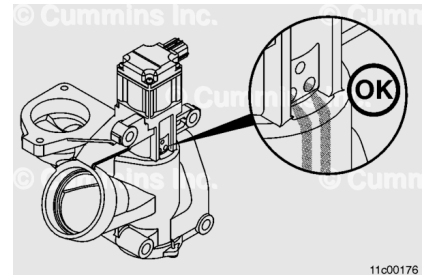
Check for soot streaks from the EGR valve gasket and seal rings.

Inspect for signs of leakage such as soot streaks. If signs of leakage are present, replace the EGR valve gaskets. Refer to Procedure 011-022 in Section 11 (</qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html>).



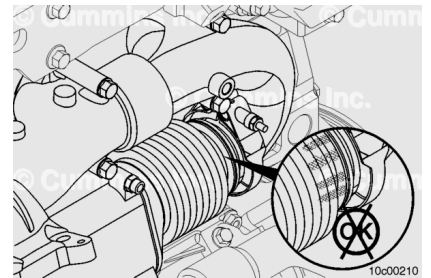
Check for leakage from the EGR valve weep hole.

Soot leakage at the EGR valve weep hole is normal and can be expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep hole.



Check for leakage from the EGR cooler bellows.

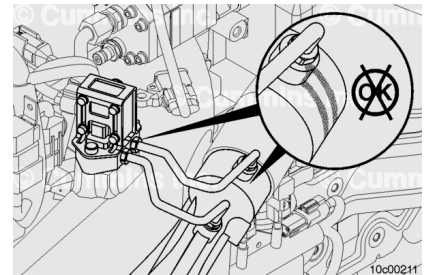
Inspect for signs of leakage such as soot streaks. If signs of leakage are present, replace the EGR cooler or gasket. Refer to Procedure 011-019 in Section 11 (</qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html>).



Check for soot leakage at the EGR differential pressure sensing tubes.

Inspect for signs of leakage such as soot streaks, appearing yellowish brown or black.

If signs of leakage are present, replace the EGR differential pressure tubes. Refer to Procedure 011-026 in Section 11 (</qs3/pubsys2/xml/en/procedures/10/10-011-026-tr.html>).



Check for soot leakage on the exhaust adapter pipe.

Inspect for signs for leakage such as soot streaks , appearing yellowish brown or black.

If signs of leakage are present, replace the gasket and V-band clamp. Refer to Procedure 011-043 in Section 11 (</qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html>).

Clean all soot stains and paint the components in order to detect future leaks.

